CASO DE ESTUDIO

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CCNA PRIMER MODULO

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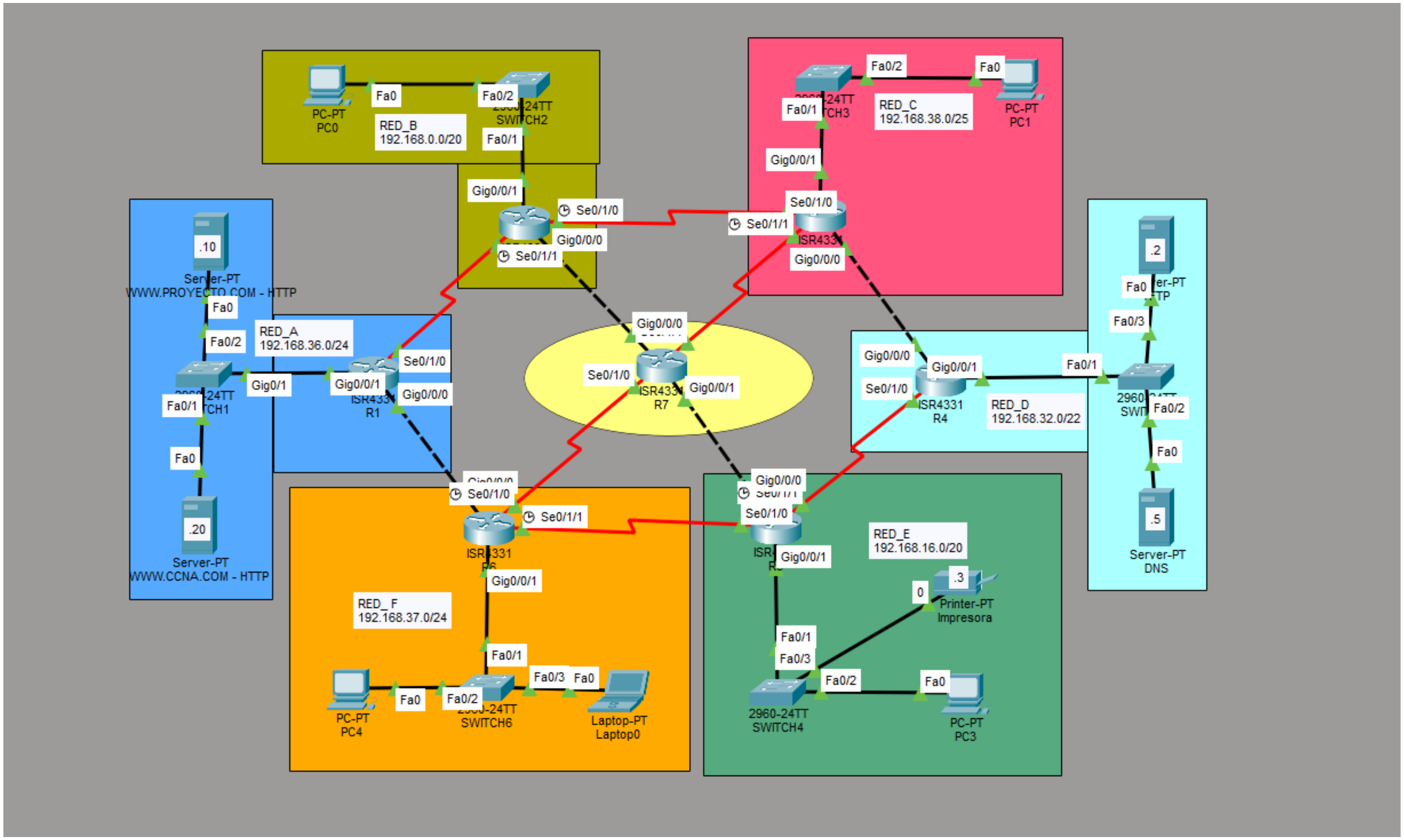
DIAGRAMA DE TOPOLOGÍA



TABLAS

Hosts Solicitados	Dirección de Red	Máscara de Subred	Máscara Wildcard	Primera IP Útil	Última IP Útil	Dirección de Broadcast
3.000	192.168.0.0	255.255.240.0	0.0.15.255	192.168.0.1	192.168.15.254	192.168.15.255
2.047	192.168.16.0	255.255.240.0	0.0.15.255	192.168.16.1	192.168.31.254	192.168.31.255
530	192.168.32.0	255.255.252.0	0.0.3.255	192.168.32.1	192.168.35.254	192.168.35.255
200	192.168.36.0	255.255.255.0	0.0.0.255	192.168.36.1	192.168.36.254	192.168.36.255
127	192.168.37.0	255.255.255.0	0.0.0.255	192.168.37.1	192.168.37.254	192.168.37.255
64	192.168.38.0	255.255.255.128	0.0.0.127	192.168.38.1	192.168.38.126	192.168.38.127

Dispositivo	Interface	Red	Dirección IP	Máscara Subred	Gateway Por defecto
Router 1	Gig0/0/0	10.0.0.20/30	10.0.0.21	255.255.255.252	-
	Gig0/0/1	192.168.36.0/24	192.168.36.1	255.255.255.0	-
	Se0/1/0	10.0.0.0/30	10.0.0.1	255.255.255.252	-
Router 2	Gig0/0/0	10.0.0.12/30	10.0.0.13	255.255.255.252	-
	Gig0/0/1	192.168.0.0/20	192.168.0.1	255.255.240.0	-
	Se0/1/0	10.0.0.4/30	10.0.0.5	255.255.255.252	-
	Se0/1/1	10.0.0.0/30	10.0.0.2	255.255.255.252	-
Router 3	Gig0/0/0	10.0.0.8/30	10.0.0.9	255.255.255.252	-
	Gig0/0/1	192.168.38.0/25	192.168.38.1	255.255.255.128	-
	Se0/1/0	10.0.0.4/30	10.0.0.6	255.255.255.252	-
	Se0/1/1	10.0.0.16/30	10.0.0.18	255.255.255.252	-
Router 4	Gig0/0/0	10.0.0.8/30	10.0.0.10	255.255.255.252	-
	Gig0/0/1	192.168.32.0/22	192.168.32.1	255.255.252.0	-
	Se0/1/0	10.0.0.32/30	10.0.0.34	255.255.255.252	-

Dispositivo	Interface	Red	Dirección IP	Máscara Subred	Gateway Por defecto
Router 5	Gig0/0/0	10.0.0.28/30	10.0.0.30	255.255.255.252	-
	Gig0/0/1	192.168.16.0/20	192.168.16.1	255.255.240.0	-
	Se0/1/0	10.0.0.36/30	10.0.0.38	255.255.255.252	-
	Se0/1/1	10.0.0.32/30	10.0.0.33	255.255.255.252	
Router 6	Gig0/0/0	10.0.0.20/30	10.0.0.22	255.255.255.252	-
	Gig0/0/1	192.168.37.0/24	192.168.37.1	255.255.255.0	-
	Se0/1/0	10.0.0.24/30	10.0.0.25	255.255.255.252	-
	Se0/1/1	10.0.0.36/30	10.0.0.37	255.255.255.252	-
Router 7	Gig0/0/0	10.0.0.12/30	10.0.0.14	255.255.255.252	-
	Gig0/0/1	10.0.0.28/30	10.0.0.29	255.255.255.252	-
	Se0/1/0	10.0.0.24/30	10.0.0.26	255.255.255.252	-
	Se0/1/1	10.0.0.16/30	10.0.0.17	255.255.255.252	-

Dispositivo	Interface	Red	Dirección IP	Máscara Subred	Gateway Por defecto
HTTP PROY	Fa0/2	192.168.36.0/24	192.168.36.10	255.255.255.0	192.168.36.1
HTTP CCNA	Fa0/1	192.168.36.0/24	192.168.36.20	255.255.255.0	192.168.36.1
PC0	Fa0/2	192.168.0.0/20	192.168.0.2	255.255.240.0	192.168.0.1
PC1	Fa0/2	192.168.38.0/25	192.168.38.2	255.255.255.128	192.168.38.1
TFTP	Fa0/3	192.168.32.0/22	192.168.32.2	255.255.252.0	192.168.32.1
DNS	Fa0/2	192.168.32.0/22	192.168.32.5	255.255.252.0	192.168.32.1
PC03	Fa0/2	192.168.16.0/20	192.168.16.2	255.255.240.0	192.168.16.1
IMPRESORA	Fa0/3	192.168.16.0/20	192.168.16.3	255.255.240.0	192.168.16.1
LAPTOP0	Fa0/3	192.168.37.0/24	192.168.37.3	255.255.255.0	192.168.37.1
PC4	Fa0/2	192.168.37.0/24	192.168.37.2	255.255.255.0	192.168.37.1

Router	Red de Destino	Máscara de Subred	Siguiente Salto	Comando CLI Completo
R1	10.0.0.0	255.255.255.252	10.0.0.2	`ip route 10.0.0.0 255.255.255.252 10.0.0.2`
R1	10.0.0.20	255.255.255.252	10.0.0.22	`ip route 10.0.0.20 255.255.255.252 10.0.0.22`
R1	192.168.0.0	255.255.240.0	10.0.0.2	`ip route 192.168.0.0 255.255.240.0 10.0.0.2`
R1	192.168.16.0	255.255.240.0	10.0.0.22	`ip route 192.168.16.0 255.255.240.0 10.0.0.22`
R1	192.168.32.0	255.255.252.0	10.0.0.2	`ip route 192.168.32.0 255.255.252.0 10.0.0.2`
R1	192.168.32.0	255.255.252.0	10.0.0.22	`ip route 192.168.32.0 255.255.252.0 10.0.0.22`
R1	192.168.37.0	255.255.255.0	10.0.0.22	`ip route 192.168.37.0 255.255.255.0 10.0.0.22`
R1	192.168.38.0	255.255.255.128	10.0.0.2	`ip route 192.168.38.0 255.255.255.128 10.0.0.2`
R2	10.0.0.0	255.255.255.252	10.0.0.1	`ip route 10.0.0.0 255.255.255.252 10.0.0.1`
R2	10.0.0.12	255.255.255.252	10.0.0.14	`ip route 10.0.0.12 255.255.255.252 10.0.0.14`
R2	10.0.0.16	255.255.255.252	10.0.0.18	`ip route 10.0.0.16 255.255.255.252 10.0.0.18`
R2	192.168.16.0	255.255.240.0	10.0.0.6	`ip route 192.168.16.0 255.255.240.0 10.0.0.6`
R2	192.168.32.0	255.255.252.0	10.0.0.6	`ip route 192.168.32.0 255.255.252.0 10.0.0.6`
R2	192.168.36.0	255.255.255.0	10.0.0.1	`ip route 192.168.36.0 255.255.255.0 10.0.0.1`
R2	192.168.37.0	255.255.255.0	10.0.0.1	`ip route 192.168.37.0 255.255.255.0 10.0.0.1`
R2	192.168.38.0	255.255.255.128	10.0.0.14	`ip route 192.168.38.0 255.255.255.128 10.0.0.14`
R2	192.168.38.0	255.255.255.128	10.0.0.6	`ip route 192.168.38.0 255.255.255.128 10.0.0.6`
R3	10.0.0.0	255.255.255.252	10.0.0.5	`ip route 10.0.0.0 255.255.255.252 10.0.0.5`
R3	10.0.0.4	255.255.255.252	10.0.0.5	`ip route 10.0.0.4 255.255.255.252 10.0.0.5`

R3	10.0.0.8	255.255.255.252	10.0.0.10	ip route 10.0.0.8 255.255.255.252 10.0.0.10
R3	10.0.0.16	255.255.255.252	10.0.0.17	ip route 10.0.0.16 255.255.255.252 10.0.0.17
R3	192.168.0.0	255.255.240.0	10.0.0.5	ip route 192.168.0.0 255.255.240.0 10.0.0.5
R3	192.168.16.0	255.255.240.0	10.0.0.17	ip route 192.168.16.0 255.255.240.0 10.0.0.17
R3	192.168.32.0	255.255.252.0	10.0.0.10	ip route 192.168.32.0 255.255.252.0 10.0.0.10
R3	192.168.36.0	255.255.255.0	10.0.0.5	ip route 192.168.36.0 255.255.255.0 10.0.0.5
R3	192.168.37.0	255.255.255.0	10.0.0.17	ip route 192.168.37.0 255.255.255.0 10.0.0.17
R4	10.0.0.0	255.255.255.252	10.0.0.9	ip route 10.0.0.0 255.255.255.252 10.0.0.9
R4	10.0.0.4	255.255.255.252	10.0.0.9	ip route 10.0.0.4 255.255.255.252 10.0.0.9
R4	10.0.0.20	255.255.255.252	10.0.0.33	ip route 10.0.0.20 255.255.255.252 10.0.0.33
R4	10.0.0.36	255.255.255.252	10.0.0.33	ip route 10.0.0.36 255.255.255.252 10.0.0.33
R4	192.168.0.0	255.255.240.0	10.0.0.9	ip route 192.168.0.0 255.255.240.0 10.0.0.9
R4	192.168.16.0	255.255.240.0	10.0.0.33	ip route 192.168.16.0 255.255.240.0 10.0.0.33
R4	192.168.36.0	255.255.255.0	10.0.0.9	ip route 192.168.36.0 255.255.255.0 10.0.0.9
R4	192.168.36.0	255.255.255.0	10.0.0.33	ip route 192.168.36.0 255.255.255.0 10.0.0.33
R4	192.168.37.0	255.255.255.0	10.0.0.33	ip route 192.168.37.0 255.255.255.0 10.0.0.33
R4	192.168.38.0	255.255.255.128	10.0.0.9	ip route 192.168.38.0 255.255.255.128 10.0.0.9
R5	10.0.0.0	255.255.255.252	10.0.0.37	ip route 10.0.0.0 255.255.255.252 10.0.0.37
R5	10.0.0.20	255.255.255.252	10.0.0.37	ip route 10.0.0.20 255.255.255.252 10.0.0.37
R5	10.0.0.24	255.255.255.252	10.0.0.27	ip route 10.0.0.24 255.255.255.252 10.0.0.27

R5	10.0.0.28	255.255.255.252	10.0.0.29	ip route 10.0.0.28 255.255.255.252 10.0.0.29
R5	10.0.0.32	255.255.255.252	10.0.0.34	ip route 10.0.0.32 255.255.255.252 10.0.0.34
R5	192.168.32.0	255.255.252.0	10.0.0.34	ip route 192.168.32.0 255.255.252.0 10.0.0.34
R5	192.168.36.0	255.255.255.0	10.0.0.37	ip route 192.168.36.0 255.255.255.0 10.0.0.37
R5	192.168.37.0	255.255.255.0	10.0.0.37	ip route 192.168.37.0 255.255.255.0 10.0.0.37
R5	192.168.38.0	255.255.255.128	10.0.0.34	ip route 192.168.38.0 255.255.255.128 10.0.0.34
R6	10.0.0.0	255.255.255.252	10.0.0.1	ip route 10.0.0.0 255.255.255.252 10.0.0.1
R6	10.0.0.24	255.255.255.252	10.0.0.26	ip route 10.0.0.24 255.255.255.252 10.0.0.26
R6	10.0.0.40	255.255.255.252	10.0.0.38	ip route 10.0.0.40 255.255.255.252 10.0.0.38
R6	192.168.0.0	255.255.240.0	10.0.0.29	ip route 192.168.0.0 255.255.240.0 10.0.0.29
R6	192.168.0.0	255.255.240.0	10.0.0.21	ip route 192.168.0.0 255.255.240.0 10.0.0.21
R6	192.168.16.0	255.255.240.0	10.0.0.38	ip route 192.168.16.0 255.255.240.0 10.0.0.38
R6	192.168.32.0	255.255.252.0	10.0.0.38	ip route 192.168.32.0 255.255.252.0 10.0.0.38
R6	192.168.36.0	255.255.255.0	10.0.0.21	ip route 192.168.36.0 255.255.255.0 10.0.0.21
R6	192.168.38.0	255.255.255.128	10.0.0.21	ip route 192.168.38.0 255.255.255.128 10.0.0.21
R7	10.0.0.0	255.255.255.252	10.0.0.13	ip route 10.0.0.0 255.255.255.252 10.0.0.13
R7	10.0.0.12	255.255.255.252	10.0.0.13	ip route 10.0.0.12 255.255.255.252 10.0.0.13
R7	10.0.0.16	255.255.255.252	10.0.0.9	ip route 10.0.0.16 255.255.255.252 10.0.0.9
R7	10.0.0.16	255.255.255.252	10.0.0.18	ip route 10.0.0.16 255.255.255.252 10.0.0.18
R7	10.0.0.24	255.255.255.252	10.0.0.25	ip route 10.0.0.24 255.255.255.252 10.0.0.25

R7	10.0.0.28	255.255.255.252	10.0.0.30	`ip route 10.0.0.28 255.255.255.252 10.0.0.30`
R7	192.168.0.0	255.255.240.0	10.0.0.13	`ip route 192.168.0.0 255.255.240.0 10.0.0.13`
R7	192.168.16.0	255.255.240.0	10.0.0.25	`ip route 192.168.16.0 255.255.240.0 10.0.0.25`
R7	192.168.32.0	255.255.252.0	10.0.0.18	`ip route 192.168.32.0 255.255.252.0 10.0.0.18`
R7	192.168.37.0	255.255.255.0	10.0.0.25	`ip route 192.168.37.0 255.255.255.0 10.0.0.25`
R7	192.168.38.0	255.255.255.128	10.0.0.18	`ip route 192.168.38.0 255.255.255.128 10.0.0.18`
R7	192.168.38.0	255.255.255.128	10.0.0.30	`ip route 192.168.38.0 255.255.255.128 10.0.0.30`

VERIFICACION DE COMANDOS:

¿Que hace cada comando?

1. **SHOW IP INTERFACE BRIEF**

Resume rápidamente el estado y configuración de todas las interfaces del dispositivo.

Es el primer comando para diagnósticos básicos.

2. **SHOW RUNNING-CONFIG**

Muestra la configuración que se está ejecutando justo en el mismo momento en la memoria RAM.

3. **SHOW IP ROUTE**

Muestra la tabla de enrutamiento IP del router, permite ver la lista de rutas que se configuran.

4. **SHOW CDP NEIGHBOR**

Muestra los dispositivos Cisco que estén conectados directamente al equipo utilizando el protocolo CDP.

5. **SHOW IP DHCP BINDING**

Muestra la lista de asignaciones de direcciones IP que el router entrega a los hosts por medio del protocolo DHCP.

ROUTER 1:

R1#SHOW IP INTERFACE BRIEF

```
R1#Show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0/0   10.0.0.21       YES manual up                    up
GigabitEthernet0/0/1   192.168.36.1    YES manual up                    up
GigabitEthernet0/0/2   unassigned      YES NVRAM  administratively down down
Serial0/1/0            10.0.0.1        YES manual up                    up
Serial0/1/1            unassigned      YES NVRAM  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
R1#
```

R1#SHOW RUNNING-CONFIG

Building configuration...

Current configuration : 1588 bytes
!
version 15.4
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname R1
!
!
!
enable secret 5 1mERr$CunnQjkAEasAf02ni13tK/
!
!
ip dhcp excluded-address 192.168.36.10

```
ip dhcp excluded-address 192.168.36.20
!
ip dhcp pool LAN
network 192.168.36.0 255.255.255.0
default-router 192.168.36.1
dns-server 192.168.32.5
!
!
!
no ip cef
no ipv6 cef
!
!
!
!
!
!
!
!
!
!
!
!
!
!
spanning-tree mode pvst
!
!
!
!
!
!
```

```
interface GigabitEthernet0/0/0
description INTERFAZ AL R6 PARA IP ROUTE
ip address 10.0.0.21 255.255.255.252
duplex auto
speed auto
!
interface GigabitEthernet0/0/1
description INTERFAZ DEL GATEWAY .1
ip address 192.168.36.1 255.255.255.0
duplex auto
speed auto
!
interface GigabitEthernet0/0/2
no ip address
duplex auto
speed auto
shutdown
!
interface Serial0/1/0
description INTERFAZ AL R2 PARA IP ROUTE
ip address 10.0.0.1 255.255.255.252
!
interface Serial0/1/1
no ip address
clock rate 2000000
shutdown
!
interface Vlan1
no ip address
shutdown
!
ip classless
```

```
ip route 192.168.38.0 255.255.255.128 10.0.0.2
ip route 192.168.0.0 255.255.240.0 10.0.0.2
ip route 192.168.32.0 255.255.252.0 10.0.0.2
ip route 192.168.37.0 255.255.255.0 10.0.0.22
ip route 192.168.16.0 255.255.240.0 10.0.0.22
ip route 192.168.32.0 255.255.252.0 10.0.0.22
!
ip flow-export version 9
!
!
!
banner motd ^C ACCESO RESTRINGIDO: Solo personal autorizado. Las conexiones son monitoreadas. ^C
!
!
!
!
line con 0
password 7 0822455D0A16
login
!
line aux 0
!
line vty 0 4
password 7 0822455D0A16
login
!
!
!
End
```

R#1SHOW IP ROUTE

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C 10.0.0.0/30 is directly connected, Serial0/1/0
L 10.0.0.1/32 is directly connected, Serial0/1/0
C 10.0.0.20/30 is directly connected, GigabitEthernet0/0/0
L 10.0.0.21/32 is directly connected, GigabitEthernet0/0/0
S 192.168.0.0/20 [1/0] via 10.0.0.2
S 192.168.16.0/20 [1/0] via 10.0.0.22
S 192.168.32.0/22 [1/0] via 10.0.0.22
[1/0] via 10.0.0.2
192.168.36.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.36.0/24 is directly connected, GigabitEthernet0/0/1
L 192.168.36.1/32 is directly connected, GigabitEthernet0/0/1
S 192.168.37.0/24 [1/0] via 10.0.0.22
192.168.38.0/25 is subnetted, 1 subnets
S 192.168.38.0/25 [1/0] via 10.0.0.2

R1#SHOW IP ROUTE

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C 10.0.0.0/30 is directly connected, Serial0/1/0
L 10.0.0.1/32 is directly connected, Serial0/1/0
C 10.0.0.20/30 is directly connected, GigabitEthernet0/0/0
L 10.0.0.21/32 is directly connected, GigabitEthernet0/0/0
S 192.168.0.0/20 [1/0] via 10.0.0.2
S 192.168.16.0/20 [1/0] via 10.0.0.22
S 192.168.32.0/22 [1/0] via 10.0.0.22
[1/0] via 10.0.0.2
192.168.36.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.36.0/24 is directly connected, GigabitEthernet0/0/1
L 192.168.36.1/32 is directly connected, GigabitEthernet0/0/1
S 192.168.37.0/24 [1/0] via 10.0.0.22
192.168.38.0/25 is subnetted, 1 subnets
S 192.168.38.0/25 [1/0] via 10.0.0.2

R1#
R1#
R1#
R1#Show cdp neighbor

Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID Local Intrfce Holdtme Capability Platform Port ID
Switch Gig 0/0/1 139 S 2960 Gig 0/1
R2 Ser 0/1/0 139 R ISR4300 Ser 0/1/1
R6 Gig 0/0/0 139 R ISR4300 Gig 0/0/0

ROUTER 2:

R2#SHOW IP INTERFACE BRIEF

```
R2#Show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0/0   10.0.0.13       YES manual up                    up
GigabitEthernet0/0/1   192.168.0.1     YES manual up                    up
GigabitEthernet0/0/2   unassigned      YES NVRAM  administratively down down
Serial0/1/0            10.0.0.5        YES manual up                    up
Serial0/1/1            10.0.0.2        YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
R2#
```

R2#SHOW RUNNING-CONFIG

Building configuration...

Current configuration : 1707 bytes
!
version 15.4
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname R2
!
!
!
enable secret 5 1mERr$CunnQjkAEasAf02ni13tK/
!
!

```
!
ip dhcp pool LAN
network 192.168.0.0 255.255.240.0
default-router 192.168.0.1
dns-server 192.168.32.5
!
!
!
no ip cef
no ipv6 cef
!
!
!
!
!
!
!
!
!
!
!
!
!
!
spanning-tree mode pvst
!
!
!
!
!
!
interface GigabitEthernet0/0/0
```

```
description INTERFAZ AL R7 PARA IP ROUTE
ip address 10.0.0.13 255.255.255.252
duplex auto
speed auto
!
interface GigabitEthernet0/0/1
description INTERFAZ DEL GATEWAY .1
ip address 192.168.0.1 255.255.240.0
duplex auto
speed auto
!
interface GigabitEthernet0/0/2
no ip address
duplex auto
speed auto
shutdown
!
interface Serial0/1/0
description INTERFAZ HACIA R3
ip address 10.0.0.5 255.255.255.252
clock rate 2000000
!
interface Serial0/1/1
description INTERFAZ AL R1 PARA IP ROUTE
ip address 10.0.0.2 255.255.255.252
clock rate 2000000
!
interface Vlan1
no ip address
shutdown
!
ip classless
```

```
ip route 192.168.16.0 255.255.240.0 10.0.0.6
ip route 192.168.38.0 255.255.255.128 10.0.0.14
ip route 192.168.38.0 255.255.255.128 10.0.0.6
ip route 192.168.32.0 255.255.252.0 10.0.0.6
ip route 192.168.36.0 255.255.255.0 10.0.0.1
ip route 192.168.37.0 255.255.255.0 10.0.0.1
ip route 10.0.0.12 255.255.255.252 10.0.0.14
ip route 10.0.0.0 255.255.255.252 10.0.0.1
ip route 10.0.0.16 255.255.255.252 10.0.0.18
!
ip flow-export version 9
!
!
!
banner motd ^C ACCESO RESTRINGIDO: Solo personal autorizado. Las conexiones son monitoreadas. ^C
!
!
!
!
line con 0
password 7 0822455D0A16
login
!
line aux 0
!
line vty 0 4
password 7 0822455D0A16
login
!
!
End
R2#SHOW IP ROUTE
```

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
C 10.0.0.0/30 is directly connected, Serial0/1/1
L 10.0.0.2/32 is directly connected, Serial0/1/1
C 10.0.0.4/30 is directly connected, Serial0/1/0
L 10.0.0.5/32 is directly connected, Serial0/1/0
C 10.0.0.12/30 is directly connected, GigabitEthernet0/0/0
L 10.0.0.13/32 is directly connected, GigabitEthernet0/0/0
C 192.168.0.0/20 is directly connected, GigabitEthernet0/0/1
192.168.0.0/32 is subnetted, 1 subnets
L 192.168.0.1/32 is directly connected, GigabitEthernet0/0/1
S 192.168.16.0/20 [1/0] via 10.0.0.6
S 192.168.32.0/22 [1/0] via 10.0.0.6
S 192.168.36.0/24 [1/0] via 10.0.0.1
S 192.168.37.0/24 [1/0] via 10.0.0.1
192.168.38.0/25 is subnetted, 1 subnets
S 192.168.38.0/25 [1/0] via 10.0.0.14
[1/0] via 10.0.0.6

R2#SHOW CDP NEIGHBOR

Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID Local Intrfce Holdtme Capability Platform Port ID
Switch Gig 0/0/1 152 S 2960 Fas 0/1
R1 Ser 0/1/1 160 R ISR4300 Ser 0/1/0
R3 Ser 0/1/0 161 R ISR4300 Ser 0/1/0
R7 Gig 0/0/0 161 R ISR4300 Gig 0/0/0

R2#SHOW CDP NEIGHBOR

Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID Local Intrfce Holdtme Capability Platform Port ID
Switch Gig 0/0/1 152 S 2960 Fas 0/1
R1 Ser 0/1/1 160 R ISR4300 Ser 0/1/0
R3 Ser 0/1/0 161 R ISR4300 Ser 0/1/0
R7 Gig 0/0/0 161 R ISR4300 Gig 0/0/0
R2#
R2#Show ip dhcp binding
IP address Client-ID/ Lease expiration Type
Hardware address
192.168.0.2 00D0.FF15.D7B1 – Automatic

ROUTER 3:

R3#SHOW IP INTERFACE BRIEF

```
R3#Show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0/0   10.0.0.9        YES manual up                    up
GigabitEthernet0/0/1   192.168.38.1    YES manual up                    up
GigabitEthernet0/0/2   unassigned      YES NVRAM  administratively down down
Serial0/1/0            10.0.0.6        YES manual up                    up
Serial0/1/1            10.0.0.18       YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
R3#
```

R3#SHOW RUNNING-CONFIG

Building configuration...

Current configuration : 1699 bytes
!
version 15.4
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname R3
!
!
!
enable secret 5 1mERr$CunnQjkAEasAf02ni13tK/
!
!
!
ip dhcp pool LAN

```
network 192.168.38.0 255.255.255.128
default-router 192.168.38.1
dns-server 192.168.32.5
!
!
!
no ip cef
no ipv6 cef
!
!
!
!
!
!
!
!
!
!
!
!
!
!
spanning-tree mode pvst
!
!
!
!
!
!
interface GigabitEthernet0/0/0
description INTERFAZ AL R4 PARA IP ROUTE
ip address 10.0.0.9 255.255.255.252
```

```
duplex auto
speed auto
!
interface GigabitEthernet0/0/1
description INTERFAZ DEL GATEWAY .1
ip address 192.168.38.1 255.255.255.128
duplex auto
speed auto
!
interface GigabitEthernet0/0/2
no ip address
duplex auto
speed auto
shutdown
!
interface Serial0/1/0
description INTERFAZ AL R2 PARA IP ROUTE
ip address 10.0.0.6 255.255.255.252
!
interface Serial0/1/1
description INTERFAZ AL R3 PARA IP ROUTE
ip address 10.0.0.18 255.255.255.252
clock rate 2000000
!
interface Vlan1
no ip address
shutdown
!
ip classless
ip route 192.168.16.0 255.255.240.0 10.0.0.17
ip route 192.168.37.0 255.255.255.0 10.0.0.17
ip route 192.168.0.0 255.255.240.0 10.0.0.5
```

```
ip route 192.168.32.0 255.255.252.0 10.0.0.10
ip route 192.168.36.0 255.255.255.0 10.0.0.5
ip route 10.0.0.8 255.255.255.252 10.0.0.10
ip route 10.0.0.4 255.255.255.252 10.0.0.5
ip route 10.0.0.16 255.255.255.252 10.0.0.17
ip route 10.0.0.0 255.255.255.252 10.0.0.5
!
ip flow-export version 9
!
!
!
banner motd ^C ACCESO RESTRINGIDO: Solo personal autorizado. Las conexiones son monitoreadas. ^C
!
!
!
!
line con 0
password 7 0822455D0A16
login
!
line aux 0
!
line vty 0 4
password 7 0822455D0A16
login
!
!
!
End
```

R3#SHOW IP ROUTE

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
S 10.0.0.0/30 [1/0] via 10.0.0.5
C 10.0.0.4/30 is directly connected, Serial0/1/0
L 10.0.0.6/32 is directly connected, Serial0/1/0
C 10.0.0.8/30 is directly connected, GigabitEthernet0/0/0
L 10.0.0.9/32 is directly connected, GigabitEthernet0/0/0
C 10.0.0.16/30 is directly connected, Serial0/1/1
L 10.0.0.18/32 is directly connected, Serial0/1/1
S 192.168.0.0/20 [1/0] via 10.0.0.5
S 192.168.16.0/20 [1/0] via 10.0.0.17
S 192.168.32.0/22 [1/0] via 10.0.0.10
S 192.168.36.0/24 [1/0] via 10.0.0.5
S 192.168.37.0/24 [1/0] via 10.0.0.17
192.168.38.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.38.0/25 is directly connected, GigabitEthernet0/0/1
L 192.168.38.1/32 is directly connected, GigabitEthernet0/0/1

R3#SHOW CDP NEIGHBOR

Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID Local Intrfce Holdtme Capability Platform Port ID
Switch Gig 0/0/1 167 S 2960 Fas 0/1
R2 Ser 0/1/0 167 R ISR4300 Ser 0/1/0
R4 Gig 0/0/0 176 R ISR4300 Gig 0/0/0
R7 Ser 0/1/1 176 R ISR4300 Ser 0/1/1

R3#SHOW IP DHCP BINDING

IP address Client-ID/ Lease expiration Type
Hardware address
192.168.38.2 0002.4A73.31C2 – Automatic

ROUTER 4:

R4#SHOW IP INTERFACE BRIEF

```
R4#SHOW IP INTERFACE BRIEF
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0/0   10.0.0.10       YES manual up                    up
GigabitEthernet0/0/1   192.168.32.1    YES manual up                    up
GigabitEthernet0/0/2   unassigned      YES unset  administratively down down
Serial0/1/0            10.0.0.34       YES manual up                    up
Serial0/1/1            unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
R4#
```

R4#SHOW RUNNING-CONFIG

Building configuration...

Current configuration : 1809 bytes
!
version 15.4
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname R4
!
!
!
enable secret 5 1mERr$CunnQjkAEasAf02ni13tK/
!
!
ip dhcp excluded-address 192.168.32.2
ip dhcp excluded-address 192.168.32.5

```
!
ip dhcp pool LAN
network 192.168.32.0 255.255.252.0
default-router 192.168.32.1
dns-server 192.168.32.5
!
!
!
ip cef
no ipv6 cef
!
!
!
!
!
!
!
!
!
!
!
!
!
!
spanning-tree mode pvst
!
!
!
!
!
!
interface GigabitEthernet0/0/0
```

```
description INTERFAZ AL R4 PARA IP ROUTE
ip address 10.0.0.10 255.255.255.252
duplex auto
speed auto
!
interface GigabitEthernet0/0/1
description INTERFAZ DEL GATEWAY .1
ip address 192.168.32.1 255.255.252.0
duplex auto
speed auto
!
interface GigabitEthernet0/0/2
no ip address
duplex auto
speed auto
shutdown
!
interface Serial0/1/0
description INTERFAZ AL R4 PARA IP ROUTE
ip address 10.0.0.34 255.255.255.252
!
interface Serial0/1/1
no ip address
clock rate 2000000
shutdown
!
interface Vlan1
no ip address
shutdown
!
ip classless
ip route 192.168.38.0 255.255.255.128 10.0.0.9
```

```
ip route 192.168.36.0 255.255.255.0 10.0.0.9
ip route 192.168.0.0 255.255.240.0 10.0.0.9
ip route 192.168.37.0 255.255.255.0 10.0.0.33
ip route 192.168.16.0 255.255.240.0 10.0.0.33
ip route 192.168.36.0 255.255.255.0 10.0.0.33
ip route 10.0.0.0 255.255.255.252 10.0.0.9
ip route 10.0.0.20 255.255.255.252 10.0.0.33
ip route 10.0.0.4 255.255.255.252 10.0.0.9
ip route 10.0.0.36 255.255.255.252 10.0.0.33
ip route 10.0.0.16 255.255.255.252 10.0.0.9
!
ip flow-export version 9
!
!
!
banner motd ^C ACCESO RESTRINGIDO: Solo personal autorizado. Las conexiones son monitoreadas. ^C
!
!
!
!
line con 0
password 7 0822455D0A16
login
!
line aux 0
!
line vty 0 4
password 7 0822455D0A16
login
!
End
R4#SHOW IP ROUTE
```

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 9 subnets, 2 masks
S 10.0.0.0/30 [1/0] via 10.0.0.9
S 10.0.0.4/30 [1/0] via 10.0.0.9
C 10.0.0.8/30 is directly connected, GigabitEthernet0/0/0
L 10.0.0.10/32 is directly connected, GigabitEthernet0/0/0
S 10.0.0.16/30 [1/0] via 10.0.0.9
S 10.0.0.20/30 [1/0] via 10.0.0.33
C 10.0.0.32/30 is directly connected, Serial0/1/0
L 10.0.0.34/32 is directly connected, Serial0/1/0
S 10.0.0.36/30 [1/0] via 10.0.0.33
S 192.168.0.0/20 [1/0] via 10.0.0.9
S 192.168.16.0/20 [1/0] via 10.0.0.33
C 192.168.32.0/22 is directly connected, GigabitEthernet0/0/1
192.168.32.0/32 is subnetted, 1 subnets
L 192.168.32.1/32 is directly connected, GigabitEthernet0/0/1
S 192.168.36.0/24 [1/0] via 10.0.0.9
[1/0] via 10.0.0.33
S 192.168.37.0/24 [1/0] via 10.0.0.33
192.168.38.0/25 is subnetted, 1 subnets
S 192.168.38.0/25 [1/0] via 10.0.0.9
R4#SHOW CDP NEIGHBOR

Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID Local Intrfce Holdtme Capability Platform Port ID
Switch Gig 0/0/1 162 S 2960 Fas 0/1
R3 Gig 0/0/0 171 R ISR4300 Gig 0/0/0
R5 Ser 0/1/0 168 R ISR4300 Ser 0/1/1

ROUTER 5:

R5#SHOW IP INTERFACE BRIEF

```
R5#SHOW IP INTERFACE BRIEF
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0/0   10.0.0.30       YES manual up                    up
GigabitEthernet0/0/1   192.168.16.1    YES manual up                    up
GigabitEthernet0/0/2   unassigned      YES unset  administratively down down
Serial0/1/0            10.0.0.38       YES manual up                    up
Serial0/1/1            10.0.0.33       YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
R5#
```

R5#SHOW RUNNING-CONFIG

Building configuration...

Current configuration : 1786 bytes
!
version 15.4
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname R5
!
!
!
enable secret 5 1mERr$CunnQjkAEasAf02ni13tK/
!
!
ip dhcp excluded-address 192.168.16.3
!

```
ip dhcp pool LAN
network 192.168.16.0 255.255.240.0
default-router 192.168.16.1
dns-server 192.168.32.5
!
!
!
ip cef
no ipv6 cef
!
!
!
!
!
!
!
!
!
!
!
!
!
!
spanning-tree mode pvst
!
!
!
!
!
!
interface GigabitEthernet0/0/0
description INTERFAZ AL R7 PARA IP ROUTE
```

```
ip address 10.0.0.30 255.255.255.252
duplex auto
speed auto
!
interface GigabitEthernet0/0/1
description INTERFAZ DEL GATEWAY .1
ip address 192.168.16.1 255.255.240.0
duplex auto
speed auto
!
interface GigabitEthernet0/0/2
no ip address
duplex auto
speed auto
shutdown
!
interface Serial0/1/0
description INTERFAZ AL R6 PARA IP ROUTE
ip address 10.0.0.38 255.255.255.252
!
interface Serial0/1/1
description INTERFAZ AL R4 PARA IP ROUTE
ip address 10.0.0.33 255.255.255.252
clock rate 2000000
!
interface Vlan1
no ip address
shutdown
!
ip classless
ip route 192.168.0.0 255.255.240.0 10.0.0.29
ip route 192.168.38.0 255.255.255.128 10.0.0.34
```

```
ip route 192.168.32.0 255.255.252.0 10.0.0.34
ip route 192.168.37.0 255.255.255.0 10.0.0.37
ip route 192.168.36.0 255.255.255.0 10.0.0.37
ip route 10.0.0.32 255.255.255.252 10.0.0.34
ip route 10.0.0.28 255.255.255.252 10.0.0.29
ip route 10.0.0.24 255.255.255.252 10.0.0.27
ip route 10.0.0.0 255.255.255.252 10.0.0.37
ip route 10.0.0.20 255.255.255.252 10.0.0.37
!
ip flow-export version 9
!
!
!
banner motd ^C ACCESO RESTRINGIDO: Solo personal autorizado. Las conexiones son monitoreadas. ^C
!
!
!
!
line con 0
password 7 0822455D0A16
login
!
line aux 0
!
line vty 0 4
password 7 0822455D0A16
login
!
!
!
End
R5#SHOW IP ROUTE
```

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
S 10.0.0.0/30 [1/0] via 10.0.0.37
S 10.0.0.20/30 [1/0] via 10.0.0.37
C 10.0.0.28/30 is directly connected, GigabitEthernet0/0/0
L 10.0.0.30/32 is directly connected, GigabitEthernet0/0/0
C 10.0.0.32/30 is directly connected, Serial0/1/1
L 10.0.0.33/32 is directly connected, Serial0/1/1
C 10.0.0.36/30 is directly connected, Serial0/1/0
L 10.0.0.38/32 is directly connected, Serial0/1/0
S 192.168.0.0/20 [1/0] via 10.0.0.29
C 192.168.16.0/20 is directly connected, GigabitEthernet0/0/1
192.168.16.0/32 is subnetted, 1 subnets
L 192.168.16.1/32 is directly connected, GigabitEthernet0/0/1
S 192.168.32.0/22 [1/0] via 10.0.0.34
S 192.168.36.0/24 [1/0] via 10.0.0.37
S 192.168.37.0/24 [1/0] via 10.0.0.37
192.168.38.0/25 is subnetted, 1 subnets
S 192.168.38.0/25 [1/0] via 10.0.0.34

R5#SHOW CDP NEIGHBOR

Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID Local Intrfce Holdtme Capability Platform Port ID
Switch Gig 0/0/1 178 S 2960 Fas 0/1
R4 Ser 0/1/1 127 R ISR4300 Ser 0/1/0
R6 Ser 0/1/0 178 R ISR4300 Ser 0/1/1
R7 Gig 0/0/0 127 R ISR4300 Gig 0/0/1

R5#SHOW IP DHCP BINDING

IP address Client-ID/ Lease expiration Type
Hardware address
192.168.16.2 0060.5C66.452D – Automatic

ROUTER 6:

R6#SHOW IP INTERFACE BRIEF

```
R6#SHOW IP INTERFACE BRIEF
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0/0   10.0.0.22       YES manual up                    up
GigabitEthernet0/0/1   192.168.37.1    YES manual up                    up
GigabitEthernet0/0/2   unassigned      YES unset  administratively down down
Serial0/1/0            10.0.0.25       YES manual up                    up
Serial0/1/1            10.0.0.37       YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
R6#
```

R6#SHOW RUNNING-CONFIG

Building configuration...

Current configuration : 1675 bytes
!
version 15.4
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname R6
!
!
!
enable secret 5 1mERr$CunnQjkAEasAf02ni13tK/
!
!
!
ip dhcp pool LAN

```
network 192.168.37.0 255.255.255.0
default-router 192.168.37.1
dns-server 192.168.32.5
!
!
!
ip cef
no ipv6 cef
!
!
!
!
!
!
!
!
!
!
!
!
!
!
spanning-tree mode pvst
!
!
!
!
!
!
interface GigabitEthernet0/0/0
description INTERFAZ AL R1 PARA IP ROUTE
ip address 10.0.0.22 255.255.255.252
```

```
duplex auto
speed auto
!
interface GigabitEthernet0/0/1
description INTERFAZ DEL GATEWAY .1
ip address 192.168.37.1 255.255.255.0
duplex auto
speed auto
!
interface GigabitEthernet0/0/2
no ip address
duplex auto
speed auto
shutdown
!
interface Serial0/1/0
description INTERFAZ AL R7 PARA IP ROUTE
ip address 10.0.0.25 255.255.255.252
clock rate 2000000
!
interface Serial0/1/1
description INTERFAZ AL R5 PARA IP ROUTE
ip address 10.0.0.37 255.255.255.252
clock rate 2000000
!
interface Vlan1
no ip address
shutdown
!
ip classless
ip route 192.168.16.0 255.255.240.0 10.0.0.38
ip route 192.168.38.0 255.255.255.128 10.0.0.21
```

```
ip route 192.168.32.0 255.255.252.0 10.0.0.38
ip route 192.168.36.0 255.255.255.0 10.0.0.21
ip route 192.168.0.0 255.255.240.0 10.0.0.21
ip route 10.0.0.40 255.255.255.252 10.0.0.38
ip route 10.0.0.0 255.255.255.252 10.0.0.1
ip route 10.0.0.24 255.255.255.252 10.0.0.26
!
ip flow-export version 9
!
!
!
banner motd ^C ACCESO RESTRINGIDO: Solo personal autorizado. Las conexiones son monitoreadas. ^C
!
!
!
!
line con 0
password 7 0822455D0A16
login
!
line aux 0
!
line vty 0 4
password 7 0822455D0A16
login
!
!
!
End
```

R6#SHOW IP ROUTE

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
C 10.0.0.20/30 is directly connected, GigabitEthernet0/0/0
L 10.0.0.22/32 is directly connected, GigabitEthernet0/0/0
C 10.0.0.24/30 is directly connected, Serial0/1/0
L 10.0.0.25/32 is directly connected, Serial0/1/0
C 10.0.0.36/30 is directly connected, Serial0/1/1
L 10.0.0.37/32 is directly connected, Serial0/1/1
S 10.0.0.40/30 [1/0] via 10.0.0.38
S 192.168.0.0/20 [1/0] via 10.0.0.21
S 192.168.16.0/20 [1/0] via 10.0.0.38
S 192.168.32.0/22 [1/0] via 10.0.0.38
S 192.168.36.0/24 [1/0] via 10.0.0.21
192.168.37.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.37.0/24 is directly connected, GigabitEthernet0/0/1
L 192.168.37.1/32 is directly connected, GigabitEthernet0/0/1
192.168.38.0/25 is subnetted, 1 subnets
S 192.168.38.0/25 [1/0] via 10.0.0.21

R6#SHOW CDP NEIGHBOR

Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID Local Intrfce Holdtme Capability Platform Port ID
Switch Gig 0/0/1 121 S 2960 Fas 0/1
R1 Gig 0/0/0 129 R ISR4300 Gig 0/0/0
R5 Ser 0/1/1 128 R ISR4300 Ser 0/1/0
R7 Ser 0/1/0 130 R ISR4300 Ser 0/1/0

R6#SHOW IP DHCP BINDING

IP address Client-ID/ Lease expiration Type
Hardware address
192.168.37.2 0005.5EC7.C6CB -- Automatic
192.168.37.3 0001.C99B.7DE1 – Automatic

ROUTER 7:

R7#SHOW IP INTERFACE BRIEF

```
R7#SHOW IP INTERFACE BRIEF
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0/0   10.0.0.14       YES manual up                    up
GigabitEthernet0/0/1   10.0.0.29       YES manual up                    up
GigabitEthernet0/0/2   unassigned      YES unset  administratively down down
Serial0/1/0            10.0.0.26       YES manual up                    up
Serial0/1/1            10.0.0.17       YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
R7#
```

R7#SHOW RUNNING-CONFIG

Building configuration...

Current configuration : 1458 bytes
!
version 15.4
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname R7
!
!
!
enable secret 5 1mERr$CunnQjkAEasAf02ni13tK/
!
!
!
!

```
!
!
ip cef
no ipv6 cef
!
!
!
!
!
!
!
!
!
!
!
!
!
!
spanning-tree mode pvst
!
!
!
!
!
!
interface GigabitEthernet0/0/0
description INTERFAZ AL R2 PARA IP ROUTE
ip address 10.0.0.14 255.255.255.252
duplex auto
speed auto
!
interface GigabitEthernet0/0/1
```

```
description INTERFAZ AL R6 PARA IP ROUTE
ip address 10.0.0.29 255.255.255.252
duplex auto
speed auto
!
interface GigabitEthernet0/0/2
no ip address
duplex auto
speed auto
shutdown
!
interface Serial0/1/0
description INTERFAZ AL R6 PARA IP ROUTE
ip address 10.0.0.26 255.255.255.252
!
interface Serial0/1/1
description INTERFAZ AL R3 PARA IP ROUTE
ip address 10.0.0.17 255.255.255.252
!
interface Vlan1
no ip address
shutdown
!
router rip
!
ip classless
ip route 192.168.16.0 255.255.240.0 10.0.0.25
ip route 192.168.0.0 255.255.240.0 10.0.0.13
ip route 192.168.38.0 255.255.255.128 10.0.0.18
ip route 192.168.37.0 255.255.255.0 10.0.0.25
ip route 192.168.38.0 255.255.255.128 10.0.0.30
ip route 192.168.32.0 255.255.252.0 10.0.0.18
```

```
!
ip flow-export version 9
!
!
!
banner motd ^C ACCESO RESTRINGIDO: Solo personal autorizado. Las conexiones son monitoreadas. ^C
!
!
!
!
line con 0
password 7 0822455D0A16
login
!
line aux 0
!
line vty 0 4
password 7 0822455D0A16
login
!
!
!
End
```

R7#SHOW IP ROUTE

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
```

P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
C 10.0.0.12/30 is directly connected, GigabitEthernet0/0/0
L 10.0.0.14/32 is directly connected, GigabitEthernet0/0/0
C 10.0.0.16/30 is directly connected, Serial0/1/1
L 10.0.0.17/32 is directly connected, Serial0/1/1
C 10.0.0.24/30 is directly connected, Serial0/1/0
L 10.0.0.26/32 is directly connected, Serial0/1/0
C 10.0.0.28/30 is directly connected, GigabitEthernet0/0/1
L 10.0.0.29/32 is directly connected, GigabitEthernet0/0/1
S 192.168.0.0/20 [1/0] via 10.0.0.13
S 192.168.16.0/20 [1/0] via 10.0.0.25
S 192.168.32.0/22 [1/0] via 10.0.0.18
S 192.168.37.0/24 [1/0] via 10.0.0.25
192.168.38.0/25 is subnetted, 1 subnets
S 192.168.38.0/25 [1/0] via 10.0.0.30
[1/0] via 10.0.0.18

R7#SHOW CDP NEIGHBOR

Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID Local Intrfce Holdtme Capability Platform Port ID
R2 Gig 0/0/0 171 R ISR4300 Gig 0/0/0
R3 Ser 0/1/1 120 R ISR4300 Ser 0/1/1
R6 Ser 0/1/0 171 R ISR4300 Ser 0/1/0
R5 Gig 0/0/1 177 R ISR4300 Gig 0/0/0